SLEEP 2026

AoU

Fitbit sleep phenotyping

Longitudinal Modeling of Sleep Timing, Duration, and Social Jetlag Across the Adult Lifespan

High-resolution wearable sleep data from the All of Us Research Program

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Support: All of Us Research Program, Office of the Director, NIH

Why Longitudinal Wearable Sleep?

Problem

- Chrono-epidemiology needs reliable long-term phenotypes.
- Self-report and short monitoring windows miss variability, seasonal structure, and work/free-day differences.
- Sleep timing is circular, behaviorally constrained, and environmentally patterned.

Aim: model age, sex, work/free-day status, and seasonality as contributors to onset, offset, midpoint, and duration.

Device-measured longitudinal sleep enables population-scale modeling of sleep timing, duration, and social jetlag.

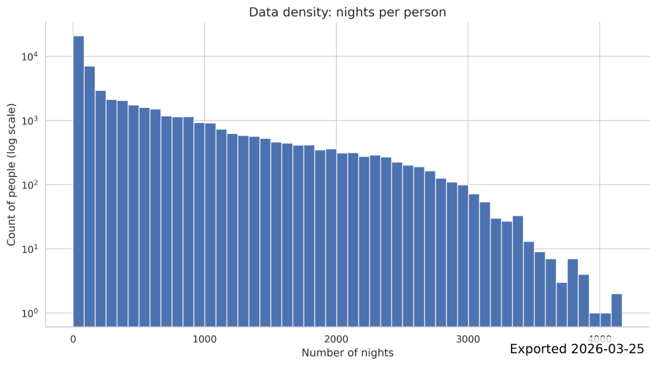
Core outputs

Onset	sleep start time
Offset	wake time
Midpoint	center of sleep episode
Duration	sleep window / time asleep

Cohort and Data Scale

- All of Us participants with filtered Fitbit sleep records.
- 22,428,610 filtered nights from 42,290 participants.
- Cohort: 71% White, 67% Female, age 52 ± 17 years.
- Median 109 nights; mean 530 nights per participant.
- Mean sleep duration 7.45h; median 7.53h.

This is a deeply longitudinal cohort, not a cross-sectional sleep snapshot.

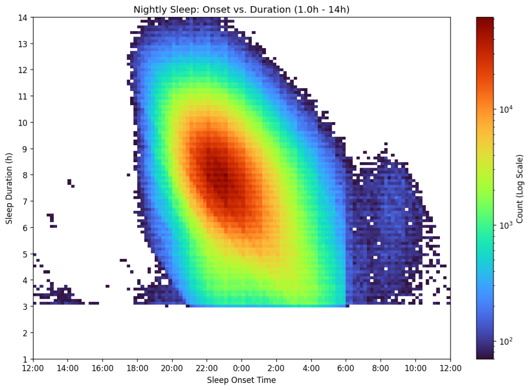


Distribution of nights per participant.

From Fitbit Records to Nightly Sleep Episodes

Filtering and episode construction

- Sleep-level records linked to daily summaries.
- Non-primary sleep excluded.
- Implausible raw durations excluded: $>$ 18h, zero/negative, common sensor artifact values.
- Segments clustered into nightly episodes; largest main sleep cluster retained per logical night.
- Sleep date assigned using a 6-hour backward shift.



Onset-duration density after filtering shows expected overnight sleep structure.

Modeling Strategy

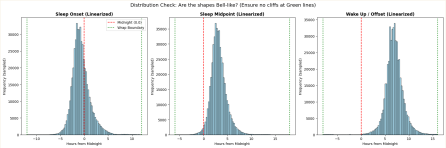
Linear mixed-effects models

$$Y_{ij} \sim \text{poly}(\text{age}_{ij}, 2) + \text{weekend}_{ij} \times \text{employment}_i + \text{sex}_i +$$

- Outcomes: onset, offset, midpoint, duration.
- Person random intercept controls stable between-person differences.
- Quadratic age captures non-linear lifespan trends.
- Employment-by-weekend interaction estimates social constraint/social jetlag patterns.

Time handling

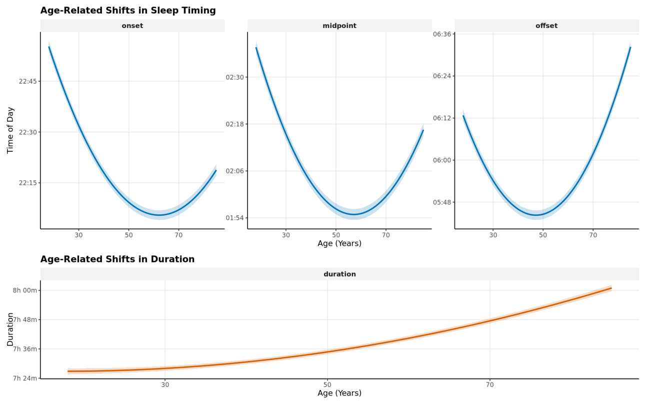
- Timing variables were linearized because distributions were primarily unimodal.
- Clock times before noon shifted forward by 24h.
- Example: 1:00 AM → 25.0; 11:00 PM → 23.0.



Result 1: Sleep Timing Advances Non-linearly With Age

- Age showed a quadratic association with sleep timing across the adult lifespan.
- Model-estimated sleep midpoint advanced by approximately 42 minutes from age 18 to age 57.
- Midpoint: 2:37 AM at age 18 vs. 1:54 AM at age 57.

A simple linear age term would miss the lifespan shape of sleep timing.

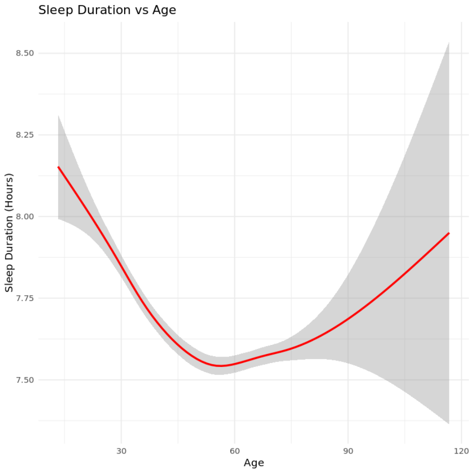


Result 2: Sleep Duration Lengthens Across Adulthood

- Duration increased with age in adjusted models.
- Estimated gain: >34 minutes between ages 18 and 85.
- Age 18: 7.45h; age 85: 8.02h.

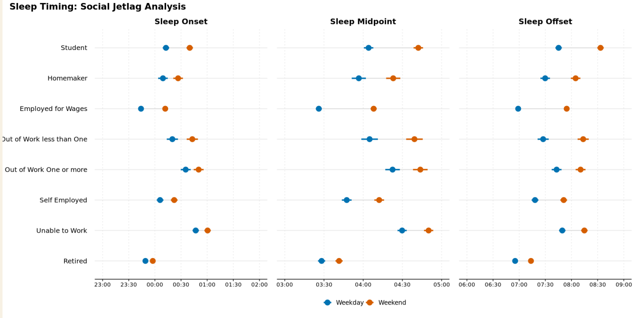
Why it matters

Timing and duration change together, so future analyses should consider mutually adjusted timing-duration phenotypes.



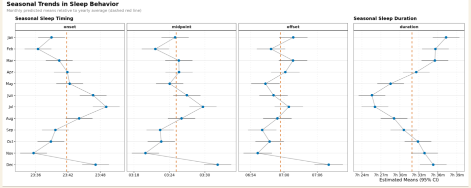
Result 3: Weekends Reveal Social Jetlag

- Weekend status delayed midpoint, onset, and offset across employment groups.
- Participants employed for wages showed the largest midpoint delay.
- Employed for wages: weekday midpoint 3:26 AM vs. weekend midpoint 4:08 AM.
- Weekend sleep duration extended by approximately 28 minutes.



Social timing constraints are visible in wearable sleep at pop-

Result 4: Sleep Is Seasonal



- Month terms indicated seasonal differences in sleep behavior.
- Sleep duration was approximately 11 minutes lower in June than the January maximum.
- Unadjusted weekly trends suggest non-sinusoidal structure, likely mixing photoperiod, calendar behavior, holidays, and DST.

What This Adds

Population-level characterization

- Quantifies non-linear age effects in device-measured sleep.
- Separates workday/free-day structure from participant-level baselines.
- Provides adjusted estimates across employment categories.

Foundation for next-stage analyses

- Genetic studies of sleep timing and chronotype.
- Clinical risk stratification using longitudinal sleep phenotypes.
- Environmental extensions: photoperiod, temperature, humidity, DST.

Large-scale wearable data can turn sleep timing from a sparse self-report phenotype into a longitudinal, model-ready exposure.

Limitations and Interpretation Guardrails

Measurement

- Fitbit-derived sleep depends on device classification of main sleep.
- Shift work is not directly measured; employment status is a proxy for social schedule constraints.
- Linearization works for the observed distribution but may under-represent true night-shift phenotypes.

Cohort and covariates

- AoU Fitbit participants are not a probability sample of U.S. adults.
- Pregnancy episode exclusion was not feasible in the V8 data context.
- ZIP3 geography is coarse for environmental and DST classification.

Conclusions

1. Sleep timing and duration are dynamically structured by non-linear age effects.
2. Employment and weekend status reveal strong social timing constraints, including measurable social jetlag.
3. Seasonal patterns persist at population scale and motivate environmental modeling.

Longitudinal wearable sleep data provide a scalable foundation for personalized circadian epidemiology.

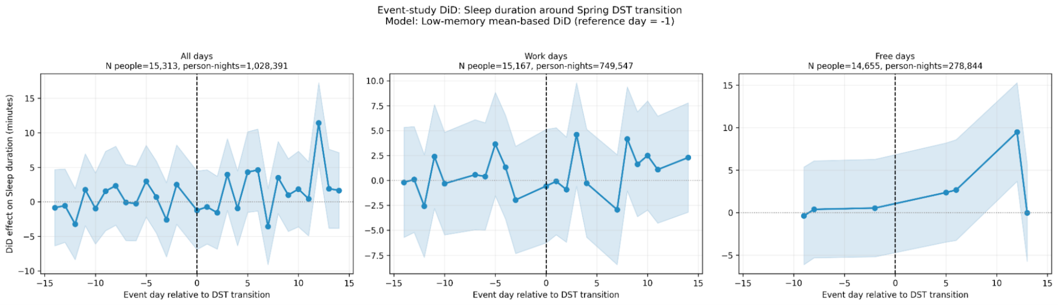
Thank you

Questions and discussion

Mason Manetta | University of California, San Francisco

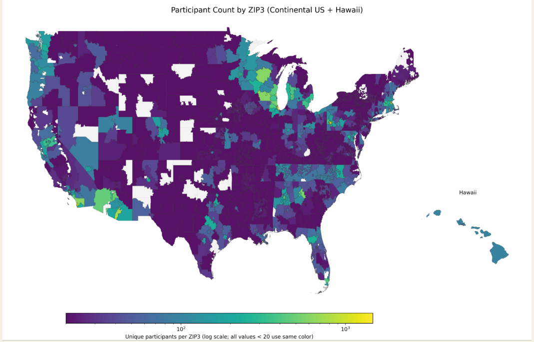
Support: All of Us Research Program, Office of the Director, NIH

Backup: Validation Against Published AoU Sleep Structure



Reference comparison with published All of Us Fitbit sleep structure; used during extraction validation.

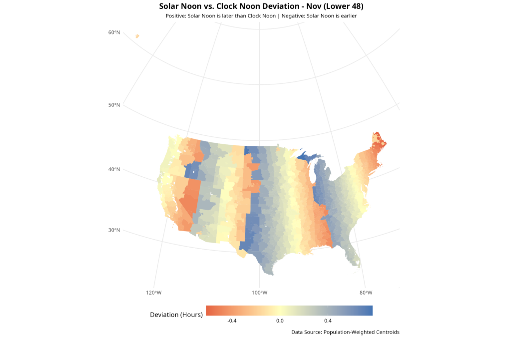
Backup: Geographic and Environmental Extensions



Participant counts by ZIP3.

- ZIP3 enables coarse geography for SES, DST, photoperiod, and PRISM weather linkage.
- Future models can estimate temperature, vapor pressure deficit, precipitation, and photoperiod effects.
- Environmental analyses should account for ZIP3 population weighting and non-random participant geography.

Backup: Daylight Saving Time Design



DST contrast

- Classify ZIP3 as DST-observing vs. non-DST control.
- Center each person-year using pre-transition baseline days -14 to -7.
- Estimate event-day differences around spring/fall transitions.